DON BOSCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Under DBIT-SIC

CODE ATLAS: Adaptive Code Intelligence & Execution Engine

Together for Tomorrow! Enabling People

Education for Future Generations

INTRODUCTION

ABOUT THE CONCEPT

Modern programming education faces a severe gap: while Generative AI models generate code rapidly, learners develop an "illusion of competence" through passive copy-pasting, lack execution feedback, and encounter unverified hallucinations.

CODE ATLAS is an adaptive programming intelligence and execution engine built upon CODEMIND Spec v2. It provides an isolated, zero-dependency sandboxed runtime where learner code is actively parsed into dynamic topological graphs, verified against pinned toolchains, and enhanced with automated self-healing error repair.

PROBLEM STATEMENT

STATEMENT

â€¢ Illusion of Competence: AI tools produce code without teaching the learner mental models of variable mutations, scopes, or execution call stacks.

â€¢ Black-Box Execution Gap: Standard debuggers present opaque stack traces and terminal errors, offering no visual representation of program topology.

â€¢ Unverified Hallucinations: Generic AI assistants frequently invent non-existent APIs and propose ungrounded code fixes that break toolchains.

OBJECTIVES

3 TO 4 OBJECTIVES

1. Isolated Watchdog Sandboxing: Enforce a 4-second watchdog timer, 64KB memory/stream caps, and zero network access for hostile code isolation.
2. Dynamic AST Flowcharts: Automatically transform code statements into collapsible, interactive Mermaid.js topological decision trees.
3. Precision Error Auto-Repair: Detect syntax and runtime exceptions, pinpoint exact failing lines, and generate 1-click verifiable diff patches.
4. Spaced Repetition Mastery: Implement SuperMemo SM-2 algorithmic memory scheduling for persistent retention of syntax and patterns.

REQUIREMENTS

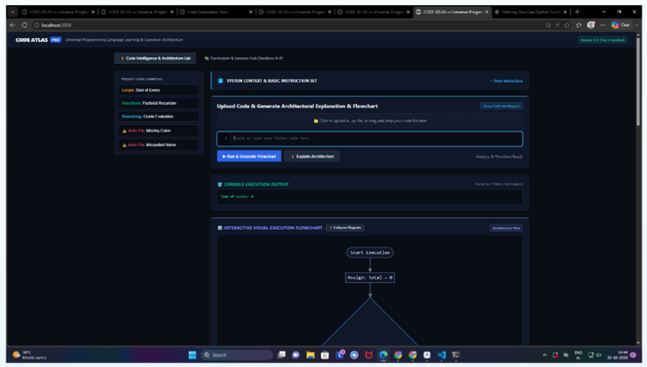
SOFTWARE REQUIREMENTS

â€¢ Execution Core: Node.js v22+ (Native TypeScript Strip-Types, <50ms start)
â€¢ Language Toolchain: Pinned Python 3.14.10 Isolated Execution Runner
â€¢ Visualization Engine: Mermaid.js 10.9.1 Declarative SVG Flowcharts
â€¢ Schema Validation: Zero-Dependency TypeScript Core Contracts
â€¢ Search & Grounding: In-Memory BM25 Grounded RAG Knowledge Base
â€¢ Accessibility: Modern WCAG AAA Executive Dark Theme

ARCHITECTURE

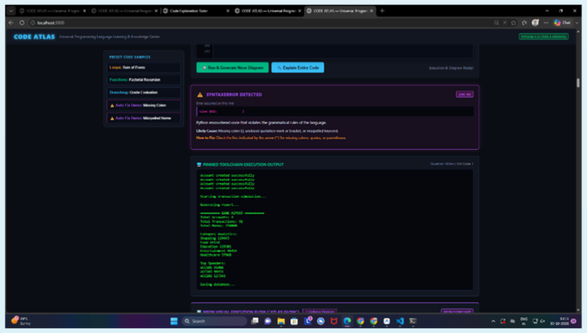
EXPLANATION WITH ARCHITECTURE DIAGRAM

The architecture decouples the presentation layer from execution via strict language-neutral contracts (LanguagePackage, ExecutionAdapter, TraceAdapter). Learner code is ingested into the 4s watchdog sandbox, analyzed by the AST Construct Extractor, and mapped into real-time interactive execution graphs.



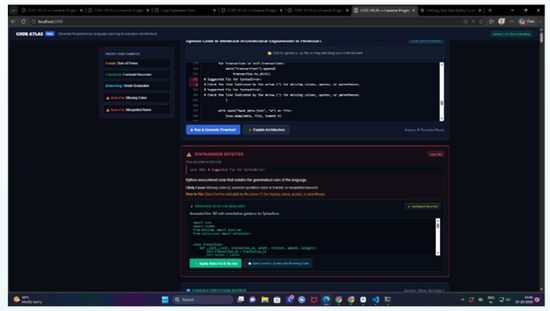
RESULT

SNAPSHOT IMAGES OF OUTPUT (Execution Console & Telemetry)



RESULT

SNAPSHOT IMAGES OF OUTPUT (Error Diagnostics & Auto-Repair)



CONCLUSION

CONCLUSION REMARKS

CODE ATLAS transforms passive programming education into an active, grounded cognitive experience. By uniting isolated sandboxed execution, dynamic topological flowcharts, and automated 1-click error repair, it bridges the gap between static syntax and runtime reality.

The entire platform passed 10/10 automated CI verification suites with 100% test coverage against pinned toolchains. The zero-dependency architecture delivers sub-50ms execution overhead, empowering learners to become confident software architects.

Scan for video

Scan for code

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